

Lab Sheet 2: Agile Working

Objectives

By completing today's workshop, you will:

- Configure a **Trello Board** specifically for your Assessment 1 (Robot System).
 - Practice "**Personal Scrum**" by adapting Agile roles for a solo developer.
 - Breakdown the Assessment Brief into **User Stories** and **Tasks**.
 - Integrate **GitHub Issues** with Trello to link your planning with your code.
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Introduction: The Solo Agile Mindset

While Scrum is typically a team sport, its principles are vital for managing individual coursework. In this module, you are wearing three hats simultaneously:

1. **The Product Owner:** You decide *what* to build (based on the Assessment Brief).
2. **The Scrum Master:** You ensure you are working efficiently and removing distractions.
3. **The Developer:** You write the code and tests.

Today, we set up the tools to manage this workflow.

Task 1: Project Board Setup

Goal: Create a Trello board tailored for the Robot Management System.

1. **Log in to Trello:** Go to trello.com (use the account created in the lecture/workshop).
2. **Create a New Board:**
 - Click **Create new board**.
 - Title: **CMP9134 - Robot Management System**.
 - Visibility: Workspace or Public (Private is fine, but you may need want to make it public to add to your assessment report for the final submission).
3. **Configure Lists for Solo Work:** Rename the standard lists to match a "Personal Kanban" flow. Create the following columns left-to-right (Click on add list):
 - **Product Backlog:** (All the ideas and requirements from the Brief).
 - **Sprint Backlog (This Week):** (What you realistically plan to finish this week).

- **In Progress (Today):** (The 1 or 2 tasks you are working on *right now*. Limit this to max 2!).
- **Waiting/Blocked:** (Tasks waiting for external help, e.g., "Ask lecturer about API").
- **Done:** (Completed tasks).

Task 2: From Assessment Brief to Backlog

Goal: Breakdown the Assessment 1 requirements into Agile User Stories.

Open the **Assessment 01 Framework Brief** (available on Blackboard). You need to extract the requirements and turn them into cards in your **Product Backlog** list.

Template for User Stories: *"As a <Role>, I want <Feature>, so that <Benefit>."*

Action: Create at least **5 cards** in your Backlog based on the brief. Use the examples below as a guide:

1. **Authentication (Security):** *"As a User, I want to log in with a secure password, so that unauthorized people cannot control the robot."*
2. **RBAC (Access Control):** *"As a 'Viewer', I want to see the robot status but NOT move it, so that I don't accidentally disrupt a mission."*
3. **Dashboard (UI):** *"As a Commander, I want to see the robot's location on a 2D grid, so that I can plan its path."*
4. **Telemetry (API Integration):** *"As a User, I want to see the current battery level, so I know when the robot needs charging."*
5. **Logging (Reliability):** *"As an Auditor, I want a log of all commands sent, so that I can investigate accidents."*

Tip: Don't forget non-functional requirements! Add cards for "Set up Docker Container", "Configure CI/CD Pipeline", and "Write AI Verification Log".

Task 3: GitHub Integration (Issues & Trello)

Goal: Link your project management (Trello) with your code repository (GitHub).

Why? In professional environments, you never write code without a ticket. Linking them allows you to click a card in Trello and see the exact Pull Request or Branch associated with it.

1. **Create your Assessment Repository if you haven't already in Week 1!).**
2. **Enable the GitHub Power-Up in Trello:**
 - On your Trello Board, click the ... **Menu** (top right) → **Power-Ups** (the small shuttle icon).
 - Search for **GitHub**.

- Click **Add** and then **Settings** to authorize your GitHub account.
- Click **Edit Power-Up Settings, Add Repo**, then link your GitHub account.
- Select your CMP9134.Assessment1 repository to link it.

3. **The Workflow (to replicate for each new issue and card pair):**

- (a) Go to your GitHub repository and create a new **Issue** (e.g., add title "Implement Authentication", description and any other details). Issues are a way to plan, discuss, and track a specific part of your work. Use the following guidance for creating the issue: <https://docs.github.com/en/issues/tracking-your-work-with-issues/learning-about-issues/quickstart>.
- (b) Fill in the description with the user story and acceptance criteria.
- (c) Save the Issue.
- (d) Open a card in Trello, or select one of the previously defined ones (e.g., the "Authentication" story).
- (e) Click the **Power-Ups** button on the bottom side of the card.
- (f) Click the **GitHub** button.
- (g) Select **Attach Issue** and select the corresponding GitHub Issue you just created.
- (h) You should now see the GitHub Issue linked to your Trello card. You can click on it to view details or navigate to GitHub directly.

4. **Linking Branches (Optional but Recommended):** When you start coding, you can create a branch specifically for that issue (e.g., `feature/auth-login`). You can attach this branch to the Trello card using the same GitHub button, giving you full traceability.

Task 4: The "Personal Scrum" Ritual

Goal: Plan your first Sprint.

You don't have a team to hold meetings with, so you must hold them with yourself.

1. **Sprint Planning:** Look at your **Product Backlog**. Move the top 3 items you want to achieve *this week* into **Sprint Backlog**. (e.g., "Set up Docker Environment" and "Basic Hello World API Connection").
2. **Daily Stand-up (Morning Routine):** Every time you sit down to work on this module:
 - Look at **In Progress**. Is it empty?
 - If yes, pull **ONE** card from **Sprint Backlog** into **In Progress**.
 - If no, finish that card before starting a new one!
3. **Definition of Done:** Before moving a card to **Done**, verify:
 - Code is written.
 - Code is pushed to GitHub.
 - (If using AI) AI Verification Log is updated.
 - It works locally.

Further Resources

- **Writing Good User Stories:** <https://www.atlassian.com/agile/project-management/user-stories>
- **Trello GitHub Power-Up Guide:** <https://support.atlassian.com/trello/docs/using-the-github-power-up/>
- **Assessment Brief:** Check Blackboard Assessment 1 folder for details regarding the Robot Management System.